PROPOSITION ML CHARACTERIZATION IN ALGEBRA.TEX

THE STACKS PROJECT AUTHORS

Editorial note. All mathematical statement and proof text is reproduced verbatim from the author's chapter algebra.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the wrapper class, the xr-hyper wiring and the \externaldocument list are omitted because the assembler supplies the class.

Editorial note. Numbering inside this standalone excerpt is editorial, not the full Stacks book print numbering.

Editorial note. The candidate is from the extractor's stacks-candidates line, not from the earlier judged initial_corpus line; the H2 judgement and its reason below were made in this module by reading the author's own statement and complete proof in the fixed checkout. Original candidate key: stacks:algebra.tex:proposition-ML-characterization.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

Source reference. Original lemma 059D: sources/algebra.tex, source lines 20973–20973. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Reading revision r002. Original author context is supplied below and editorial reference bindings are qualified by the chapter. Mathematical statement and proof are unchanged. Numbering is editorial; cited inputs remain in the complete fixed source.

Original author context (uncounted).

Definition 0.1. Let (M_i, f_{ij}) be a directed system of R-modules. We say that (M_i, f_{ij}) is a *Mittag-Leffler directed system of modules* if each M_i is an R-module of finite presentation and if for every R-module N, the inverse system

$$(\mathrm{Hom}_R(M_i, N), \mathrm{Hom}_R(f_{ij}, N))$$

is Mittag-Leffler.

We are going to characterize those R-modules that are colimits of Mittag-Leffler directed systems of modules.

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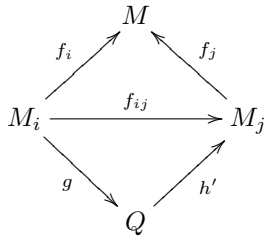
Definition 0.2. Let $f : M \to N$ and $g : M \to M'$ be maps of R-modules. Then we say g *dominates* f if for any R-module Q, we have $\mathrm{Ker}(f \otimes_R \mathrm{id}_Q) \subset \mathrm{Ker}(g \otimes_R \mathrm{id}_Q)$.

Original author statement and proof.

Proposition 0.3. *Let M be an R-module. Let (M_i, f_{ij}) be a directed system of finitely presented R-modules, indexed by I, such that $M = \mathrm{colim}\, M_i$. Let $f_i : M_i \to M$ be the canonical map. The following are equivalent:*

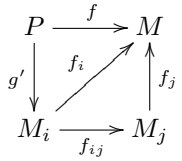
1. *For every finitely presented R-module P and module map $f : P \to M$, there exists a finitely presented R-module Q and a module map $g : P \to Q$ such that g and f dominate each other, i.e., $\mathrm{Ker}(f \otimes_R id_N) = \mathrm{Ker}(g \otimes_R id_N)$ for every R-module N.*
2. *For each $i \in I$, there exists $j \geq i$ such that $f_{ij} : M_i \to M_j$ dominates $f_i : M_i \to M$.*
3. *For each $i \in I$, there exists $j \geq i$ such that $f_{ij} : M_i \to M_j$ factors through $f_{ik} : M_i \to M_k$ for all $k \geq i$.*
4. *For every R-module N, the inverse system $(\mathrm{Hom}_R(M_i, N), \mathrm{Hom}_R(f_{ij}, N))$ is Mittag-Leffler.*
5. *For $N = \prod_{s \in I} M_s$, the inverse system $(\mathrm{Hom}_R(M_i, N), \mathrm{Hom}_R(f_{ij}, N))$ is Mittag-Leffler.*

Proof. First we prove the equivalence of (1) and (2). Suppose (1) holds and let $i \in I$. Corresponding to the map $f_i : M_i \to M$, we can choose $g : M_i \to Q$ as in (1). Since M_i and Q are of finite presentation, so is $\mathrm{Coker}(g)$. Then by Lemma 059D, $f_i : M_i \to M$ factors through $g : M_i \to Q$, say $f_i = h \circ g$ for some $h : Q \to M$. Then since Q is finitely presented, h factors through $M_j \to M$ for some $j \geq i$, say $h = f_j \circ h'$ for some $h' : Q \to M_j$. In total we have a commutative diagram



Thus f_{ij} dominates g. But g dominates f_i, so f_{ij} dominates f_i.

Conversely, suppose (2) holds. Let P be of finite presentation and $f : P \to M$ a module map. Then f factors through $f_i : M_i \to M$ for some $i \in I$, say $f = f_i \circ g'$ for some $g' : P \to M_i$. Choose by (2) a $j \geq i$ such that f_{ij} dominates f_i. We have a commutative diagram



From the diagram and the fact that f_{ij} dominates f_i, we find that f and $f_{ij} \circ g'$ dominate each other. Hence taking $g = f_{ij} \circ g' : P \to M_j$ works.

Next we prove (2) is equivalent to (3). Let $i \in I$. It is always true that f_i dominates f_{ik} for $k \geq i$, since f_i factors through f_{ik}. If (2) holds, choose $j \geq i$ such that f_{ij} dominates f_i. Then since domination is a transitive relation, f_{ij} dominates f_{ik} for $k \geq i$. All M_i are of finite presentation, so $\mathrm{Coker}(f_{ik})$ is of finite presentation for $k \geq i$. By Lemma 059D, f_{ij} factors through f_{ik} for all $k \geq i$. Thus (2) implies

(3). On the other hand, if (3) holds then for any R-module N, $f_{ij} \otimes_R \mathrm{id}_N$ factors through $f_{ik} \otimes_R \mathrm{id}_N$ for $k \geq i$. So $\mathrm{Ker}(f_{ik} \otimes_R \mathrm{id}_N) \subset \mathrm{Ker}(f_{ij} \otimes_R \mathrm{id}_N)$ for $k \geq i$. But $\mathrm{Ker}(f_i \otimes_R \mathrm{id}_N : M_i \otimes_R N \to M \otimes_R N)$ is the union of $\mathrm{Ker}(f_{ik} \otimes_R \mathrm{id}_N)$ for $k \geq i$. Thus $\mathrm{Ker}(f_i \otimes_R \mathrm{id}_N) \subset \mathrm{Ker}(f_{ij} \otimes_R \mathrm{id}_N)$ for any R-module N, which by definition means f_{ij} dominates f_i.

It is trivial that (3) implies (4) implies (5). We show (5) implies (3). Let $N = \prod_{s \in I} M_s$. If (5) holds, then given $i \in I$ choose $j \geq i$ such that

$$\mathrm{Im}(\mathrm{Hom}(M_j, N) \to \mathrm{Hom}(M_i, N)) = \mathrm{Im}(\mathrm{Hom}(M_k, N) \to \mathrm{Hom}(M_i, N))$$

for all $k \geq j$. Passing the product over $s \in I$ outside of the Hom's and looking at the maps on each component of the product, this says

$$\mathrm{Im}(\mathrm{Hom}(M_j, M_s) \to \mathrm{Hom}(M_i, M_s)) = \mathrm{Im}(\mathrm{Hom}(M_k, M_s) \to \mathrm{Hom}(M_i, M_s))$$

for all $k \geq j$ and $s \in I$. Taking $s = j$ we have

$$\mathrm{Im}(\mathrm{Hom}(M_j, M_j) \to \mathrm{Hom}(M_i, M_j)) = \mathrm{Im}(\mathrm{Hom}(M_k, M_j) \to \mathrm{Hom}(M_i, M_j))$$

for all $k \geq j$. Since f_{ij} is the image of $\mathrm{id} \in \mathrm{Hom}(M_j, M_j)$ under $\mathrm{Hom}(M_j, M_j) \to \mathrm{Hom}(M_i, M_j)$, this shows that for any $k \geq j$ there is $h \in \mathrm{Hom}(M_k, M_j)$ such that $f_{ij} = h \circ f_{ik}$. If $j \geq k$ then we can take $h = f_{kj}$. Hence (3) holds. □